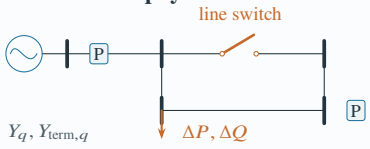
PHYSICAL ATTRIBUTION

Network physics → competing histories → time-stamped decisions

Network and physical interventions



Restoration leaves a physical response



$x(\kappa^+) = x(\kappa^-)$; instrument memory also persists

Sparse PMU observations



$|V|, \angle V, |I|, \angle I, f, \dot{f}$

8 PMUs; 47 coordinates; 30 reports/s

Oriented terminals + causal instrument model

Implemented core

new report r_k

1 Physical operators

$$\dot{x} = f_q(x, z, u)$$
$$0 = g_q(x, z, u)$$



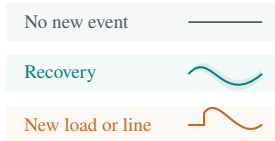
branch stamps → reduced maps

$T_o, J_o, g_o, u_o, j_o, v_o$

Prefix-fit context

Fixed network alternatives

2 History bank



$(m_h, d_h, \mu_h, v_h, \pi_h)$

State–severity dependence

3 Evidence update



$e_h = r_k - \hat{r}_{h,k}$

Joint covariance

$$L_{h,k} = \int p(r_k \mid h, a)\, p_h(a)\, da$$
$$\pi_{h,k} \propto L_{h,k}\, \pi_{h,k}^{\text{prior}}$$

Normalize; prune; retain state

4 Publish separately



M_c: active Γ_c: cause

CAL threshold + confirmation

Type / source / restoration

No reset at physical OFF

Shared covariance; prefix-fitted noise

Next report: state and history weights

ADAPTIVE DEVELOPMENT

Mechanism diagnostics; integration into the history bank remains pending

Preserve recovery uncertainty



Same mean

Different predictive spread

$$P_{\text{mix}} = \sum_h \pi_h [P_h + (\mu_h - \bar{\mu})(\mu_h - \bar{\mu})^\top]$$

Adapt noise without hiding events



Score first; update after. Keep center fixed.

Revise a bounded history window



Saved boundary; replace suffix evidence

CYBER LAYER

External methods + telemetry → Quality / availability interface → Separate alerts and validation

Drawings are schematic. Cyber scores are not extra physical evidence. Dashed modules are not enabled in the reported history-bank runs.